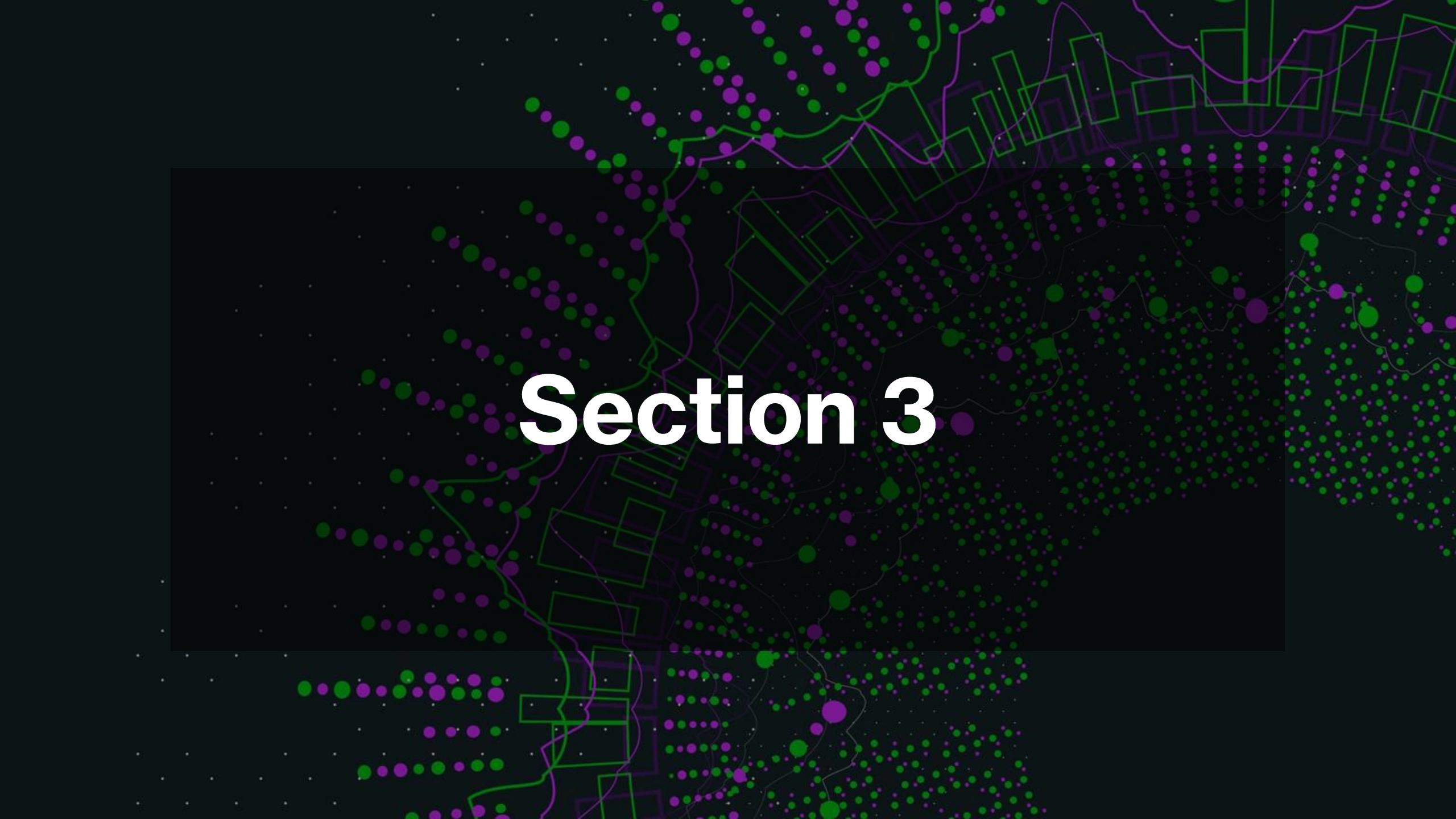
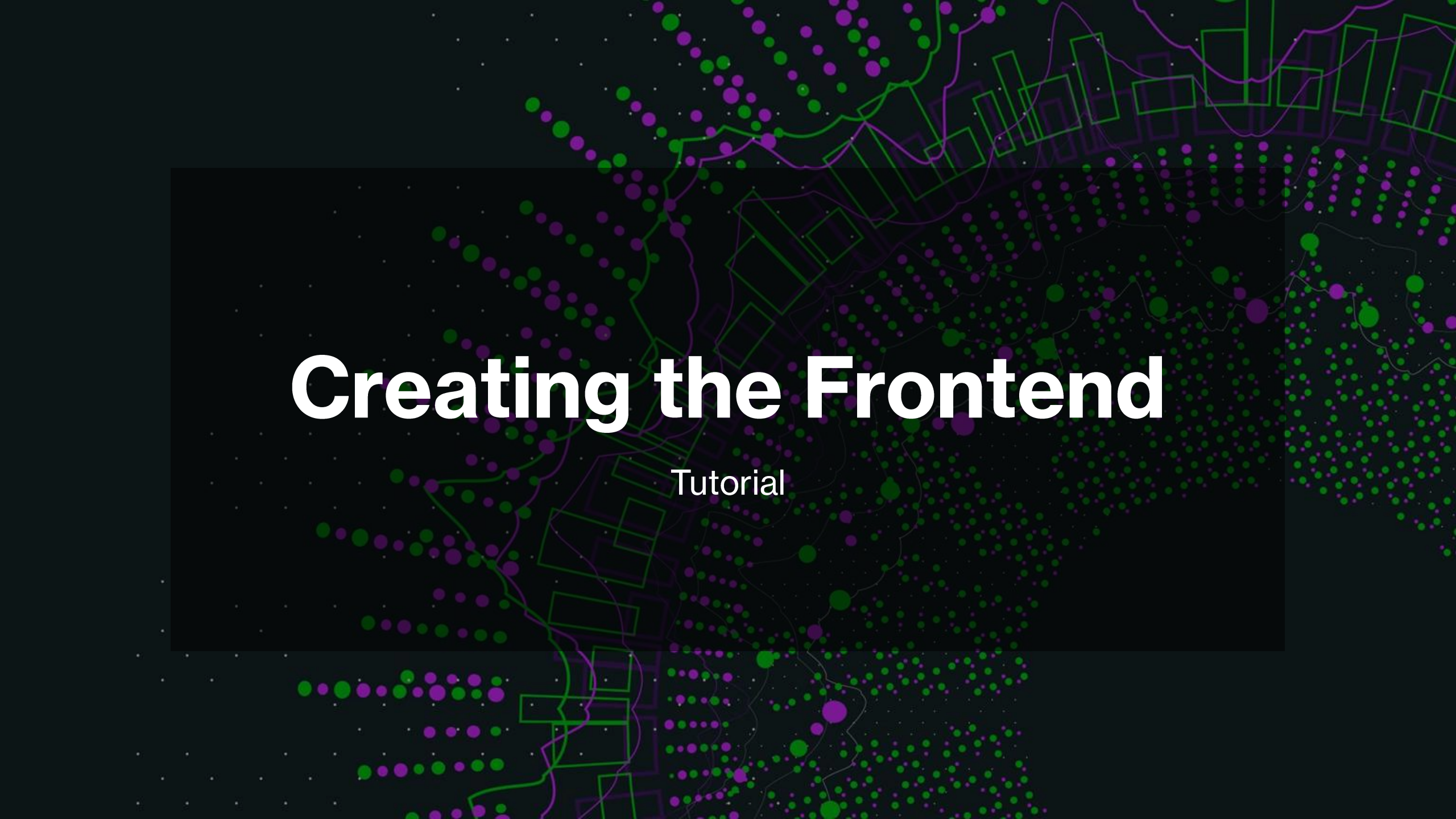




Section 3

Section Overview

- Create the frontend
- Integrate the frontend with the backend
- Create and run unit tests for the frontend
- Create and run integration tests
- Create and run end-to-end tests
- Migrate tests into GitHub Actions



Creating the Frontend

Tutorial

Functional Requirements

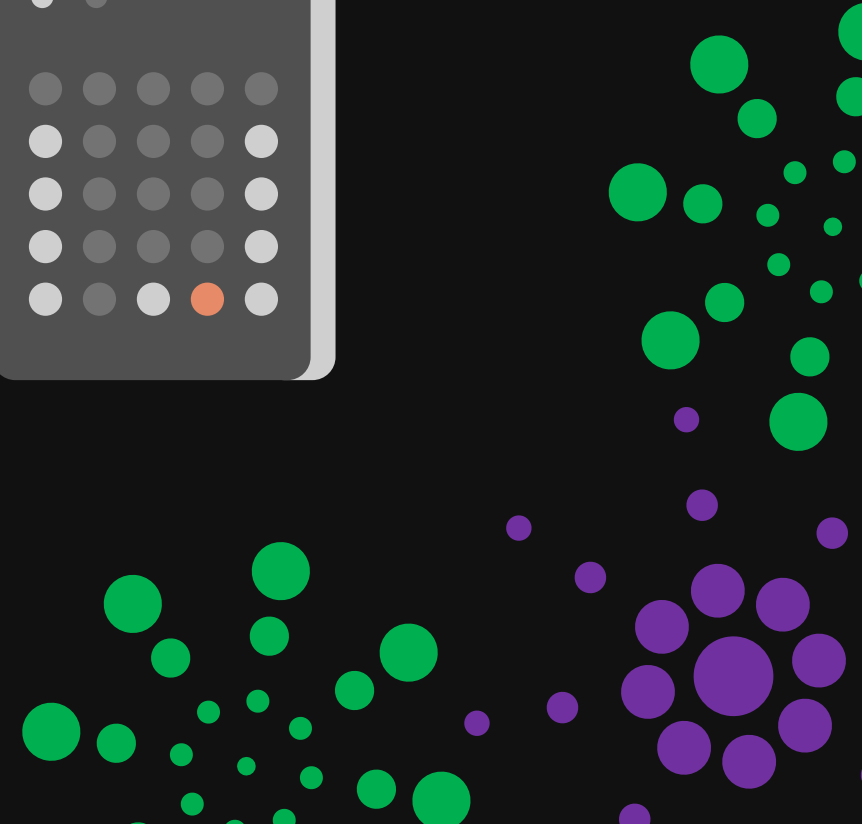
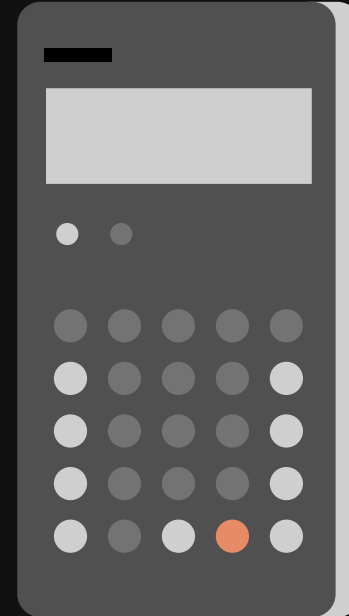
- Arithmetic Calculations:
 - Perform basic arithmetic functions:
 - Perform addition
 - Perform subtraction
 - Perform multiplication
 - Perform division
- Input Handling:
 - Accept input
 - Input must be 1 or 2 digits
 - Input must contain valid hexadecimal numbers
 - Input must include a valid operator (+, -, /, *)
- Output Handling:
 - Return and display output
 - Output must be less than 5 digits (≤ 4)
 - Output must not be negative
 - Output must not contain decimal places
- Error Handling:
 - Invalid hexadecimal input must return an error
 - Missing or invalid operator must return an error
 - Input exceeding 2 digits must return an error
 - Division by zero must return an error
 - Output that is greater than 4 digits will return an error
 - Output that is negative will return an error
 - Output with decimal places will return an error

Frontend Requirements

- Input Handling
 - Buttons for hexadecimal digits 0 to F
 - Buttons for arithmetic operators
 - An area to display what the user inputs
 - Clear Button
 - Backspace Button
- Arithmetic Calculations
 - Button for equal sign
- Output Handling
 - An area to display what the output of a calculation
- Error Handling
 - A toast to display errors
 - Validation for input
 - Validation for output

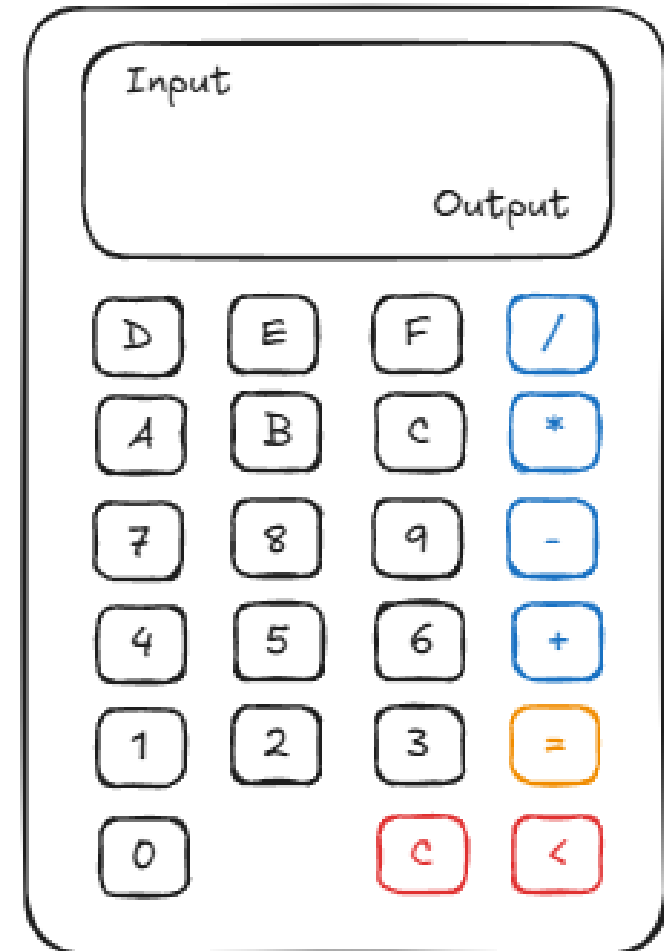
Frontend Setup

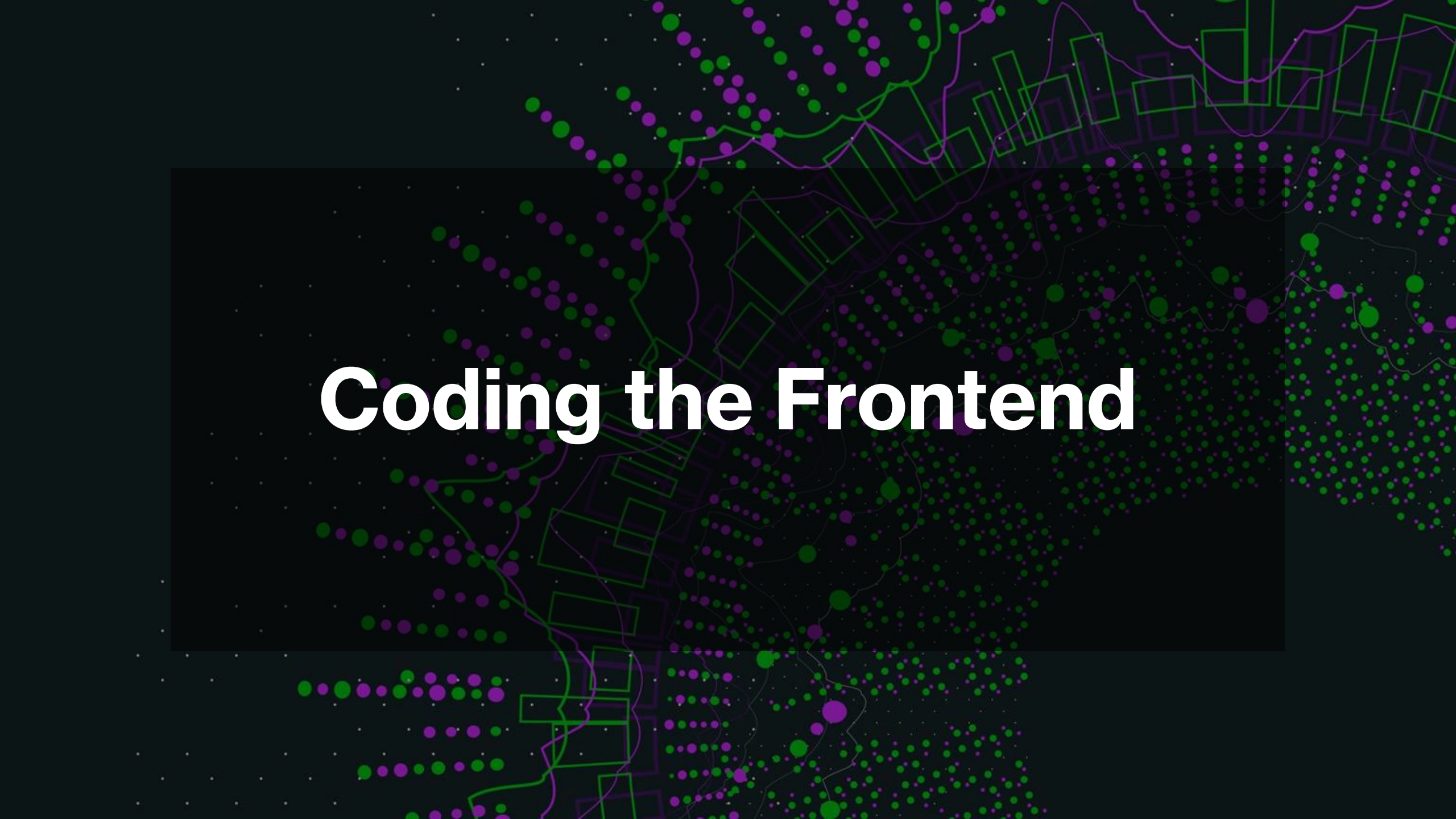
- Select a frontend stack
 - Vue.js
 - Tailwind CSS



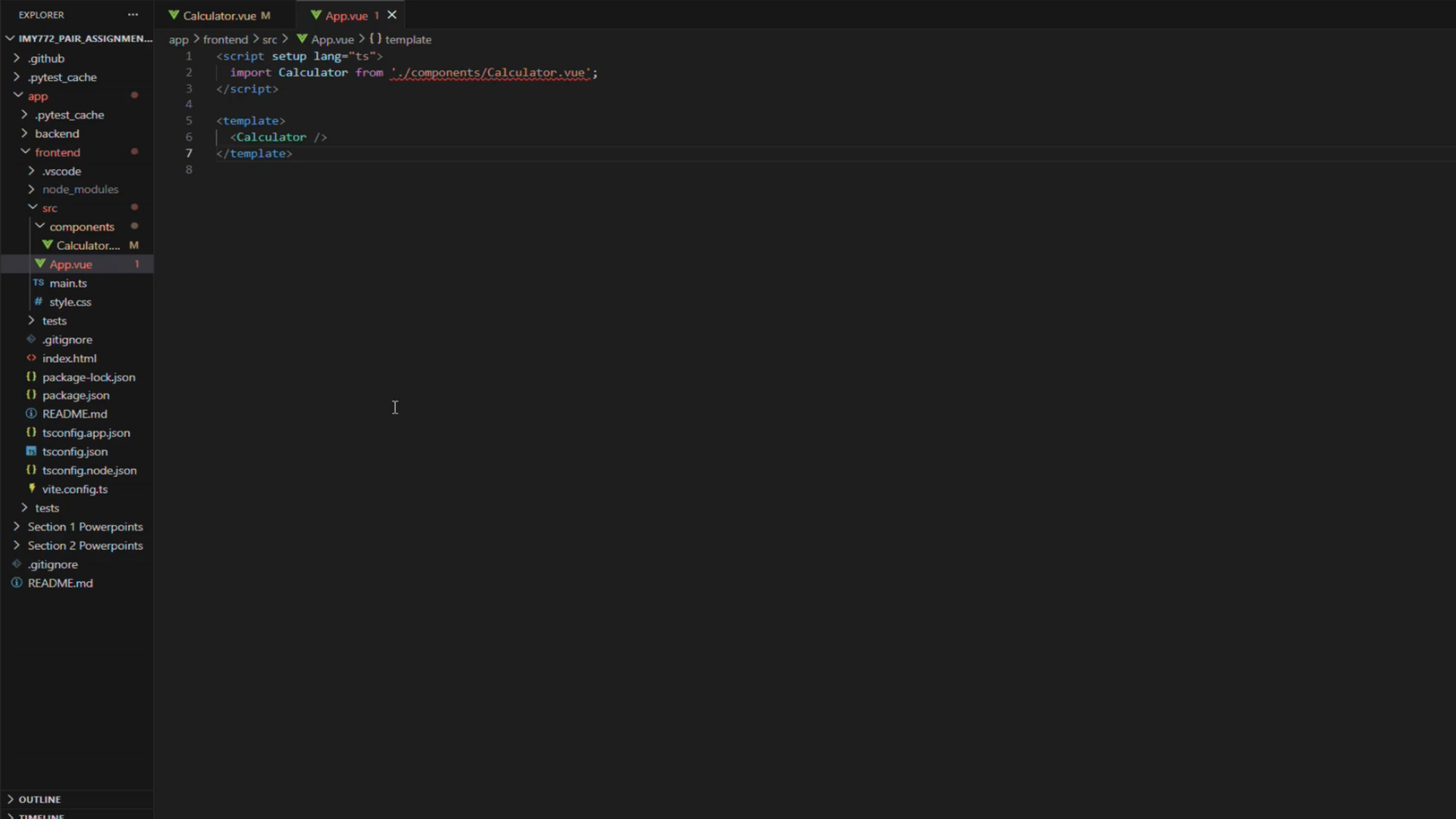
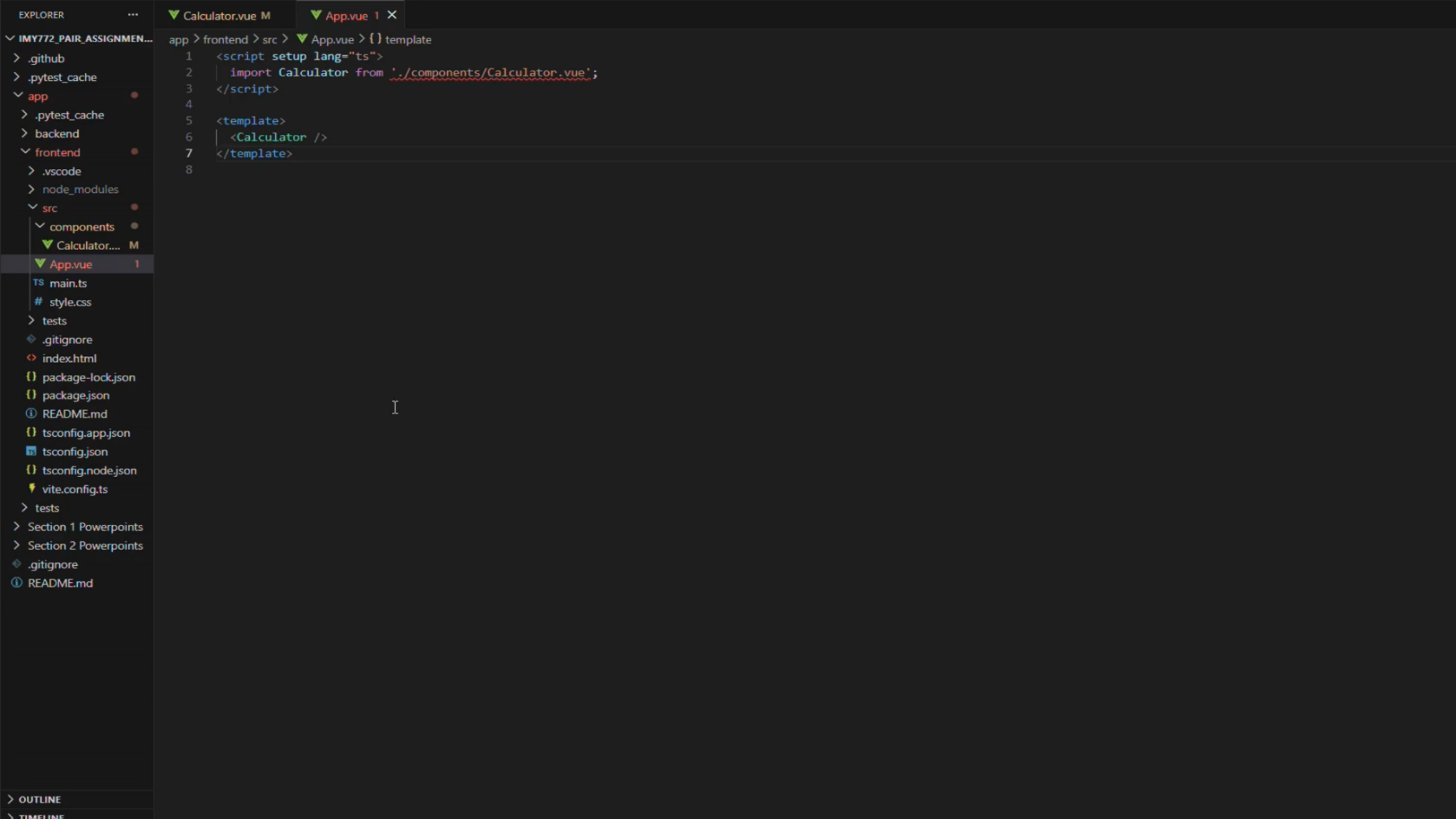
Create a wireframe

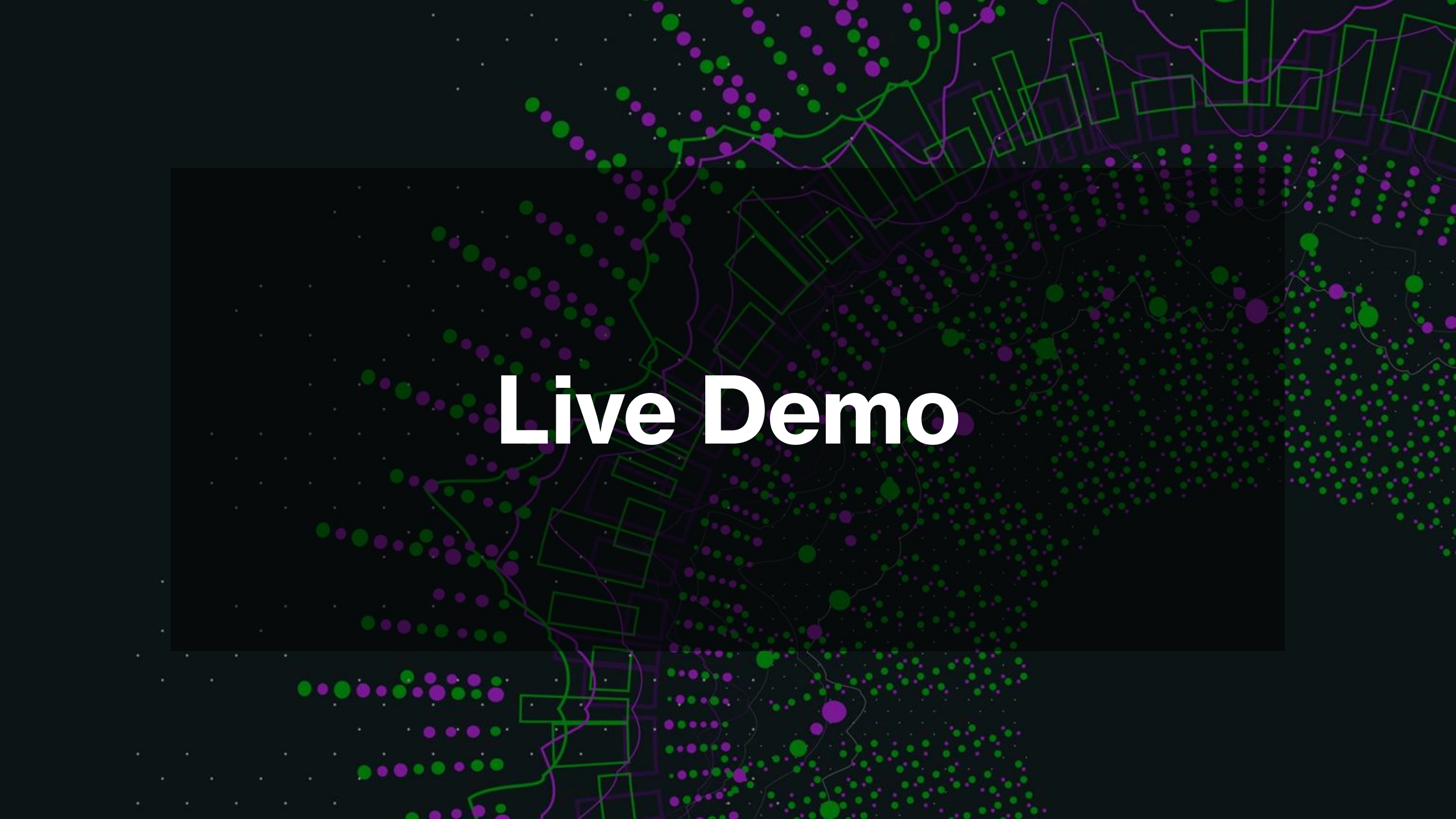
- Helps plan layout, structure, and user experience
- Tools:
 - Figma
 - Excalidraw



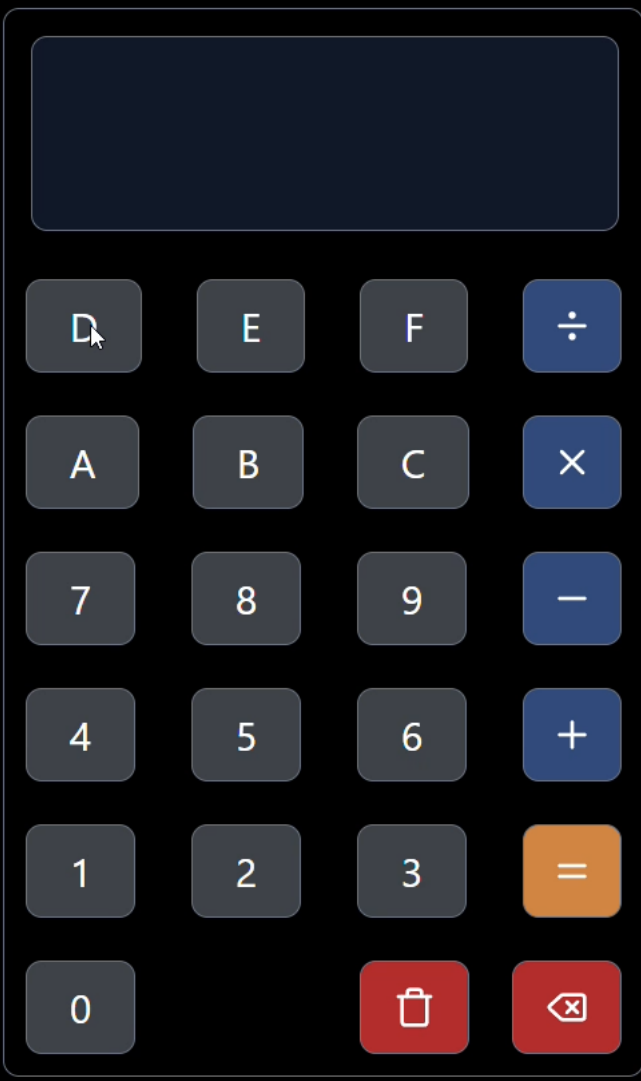


Coding the Frontend



The background is a dark gray field filled with a complex pattern of green and purple dots, lines, and rectangles. The dots are arranged in various clusters and sequences, some appearing as straight lines and others as more chaotic groups. Thin green and purple lines meander across the field, sometimes connecting the dots. Rectangles of varying sizes and orientations are scattered throughout, some appearing as simple outlines and others as more complex, overlapping shapes. The overall effect is a dense, abstract, and somewhat chaotic visual field.

Live Demo





Next Step

Integration between frontend and backend